

Automated Construction of Multi-Element MEAM Interatomic Potentials

A DFT-Informed Neural-Network Optimization Framework — Final Report

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PHYS 400 — Final Presentation





1. Motivation & engineering context

- Why elastic constants govern alloy design
- Young's modulus E for mechanical engineers
- Poisson's ratio ν for mechanical engineers
- From C_{11} , C_{12} to (E, ν)

2. The MEAM gap

- Existing libraries, coverage limits
- Pipeline overview

3. The five-stage pipeline

- Stage 1 — Configuration generator
- Stage 2 — LAMMPS molecular dynamics
- Stage 3 — Quantum Espresso DFT
- Stage 4 — MEAM initialization
- Stage 5 — Neural-network optimization

4. Results & honest discussion

- Composition surrogate (working)
- NNIP verification (work-in-progress)
- Failure analysis & outlook



- Structural alloys (Al 7075, Inconel, Ti 6Al-4V, ...) are selected on the basis of their **stiffness**, **ductility** and **Poisson response**, all of which derive from a small set of elastic constants.
- Direct experimental measurement is costly: a single single-crystal tensile + ultrasonic characterization requires weeks of sample preparation.
- Classical **molecular dynamics** (MD) can compute the full elastic tensor in minutes per composition, but *only if an accurate interatomic potential is available*.
- For metallic systems the **Modified Embedded Atom Method** (MEAM) [1, 2]¹ remains the de-facto formalism.

Objective

Build MEAM potentials *automatically* for arbitrary subsets of elements, with no manual parameter fitting and no experimental input — a closed loop from first principles to molecular dynamics.

¹MEAM = Modified Embedded Atom Method — a semi-empirical formalism that adds angular dependence to the EAM embedding function. ▶



Definition. In uniaxial loading, $E = \frac{\sigma}{\varepsilon}$ — axial stress per unit axial strain (units: GPa).

Where E appears in design:

- **Beam deflection.** Cantilever tip deflection $\delta = \frac{FL^3}{3EI}$ — a 2× change in E doubles tip rigidity for the same beam.
- **Buckling load.** Euler $P_{cr} = \frac{\pi^2 EI}{(KL)^2}$ — compression-member capacity scales linearly with E .
- **Natural frequency.** For a clamped beam $\omega_n \propto \sqrt{E/\rho}$ — vibration design, modal analysis, NVH (noise-vibration-harshness).
- **Hertz contact stress.** Bearing/gear contact area and pressure depend on the effective E^* of mating bodies.

Typical engineering ranges

Material	E (GPa)
Polymers (PE, PP)	0.5 – 3
Magnesium alloys	40 – 45
Aluminium 7075	70 – 75
Titanium 6Al-4V	110
Steel (mild)	200
Tungsten	410

E is what designers minimise when they want flexibility (springs, panels) and maximise when they want rigidity (machine frames, beams).



Definition. $\nu = -\frac{\varepsilon_{\text{lateral}}}{\varepsilon_{\text{axial}}}$ — the lateral contraction per unit axial extension (dimensionless).

Where ν appears in design:

- **Plane-strain vs plane-stress.** The transition criterion in thin- vs thick-walled pressure vessels and sheet-metal forming hinges on ν .
- **Volumetric vs deviatoric coupling.** Bulk modulus $K = \frac{E}{3(1-2\nu)}$ diverges as $\nu \rightarrow 0.5$ — incompressible limit. Critical for rubber seals, soft polymers, energy-absorbing foams (auxetic if $\nu < 0$).
- **Stress concentration.** Notch and hole stress concentration factors at finite thickness depend on ν through the Lamé parameters.
- **Fatigue & fracture.** Plane-strain K_{Ic} vs plane-stress K_c thresholds differ by a $(1 - \nu^2)$ factor.

Stability bounds (isotropic)

$$-1 < \nu < 0.5.$$

For *cubic* crystals, the Born–Cauchy criterion $C_{11} > C_{12}$ is the equivalent mechanical-stability bound; violating it gives a negative shear modulus.

Typical ranges

Material	ν
Cork (auxetic-ish)	~ 0.0
Al alloys	0.33
Steel	0.30
Rubber	$\rightarrow 0.5$
Beryllium	0.03



For a cubic crystal the engineering moduli are recovered analytically from the two independent stiffness constants:

$$E = \frac{(C_{11} - C_{12})(C_{11} + 2C_{12})}{C_{11} + C_{12}}, \quad \nu = \frac{C_{12}}{C_{11} + C_{12}}.$$

The pitfall. $\nu = C_{12}/(C_{11} + C_{12})$ is ill-conditioned — a 5 % error on each C_{ij} can blow up to a 30 % relative error on ν for Mg/Zn-rich compositions where the denominator is small.

Decision. Both ML surrogates (composition NN and the NN-MEAM optimizer) train under Huber loss on C_{11} and C_{12} , and recover (E, ν) after prediction.

Physicality filter (applied everywhere)

A configuration is dropped if any of:

- $E < 0$
- $\nu < 0$
- $\nu \geq 0.48$ (guard for $1 - 2\nu$ singularity)
- $C_{11} < C_{12}$ (Born–Cauchy violation)

The same filter is enforced in Stage 2 (LAMMPS), Stage 5a (composition NN), and Stage 5b (NN-MEAM optimizer) to keep training, validation, and verification on the same playing field.

Limitations of Existing MEAM Libraries



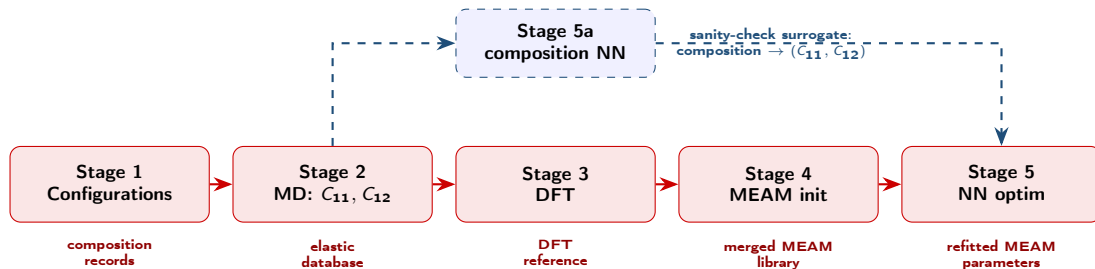
Published MEAM libraries cover *disjoint subsets* of elements; no single file spans the nine elements specified by Al7075 (Al, Cr, Cu, Fe, Mg, Mn, Si, Ti, Zn).

Library File	Al	Co	Cr	Cu	Fe	Mg	Mn	Mo	Ni	Si	Ti	Zn
AlMgZn	✓					✓						✓
AlSiCuMgFe	✓			✓	✓	✓				✓		
CuMo				✓				✓				
FeMnNiTiCuCrCoAl	✓	✓	✓	✓	✓		✓		✓		✓	
Merged (this work)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

- **Cross-interaction parameters** between elements drawn from distinct libraries are undefined; naive merging produces a potential that cannot be evaluated.
- **Purely data-driven MLIPs** [3–5]² avoid the coverage problem but require very large training corpora (10^5 DFT configurations) and forgo the physical interpretability of MEAM.
- This work fills the gap by *merging* the existing libraries and *refitting* cross-terms from DFT references.

² MLIP = Machine-Learning Interatomic Potential.

Methodology: Five-Stage Pipeline



Stage	Description	Framework	Status
1	Random alloy configuration generation	Python	Complete
2	MD elastic-property evaluation	LAMMPS Python API	Complete
3	DFT reference calculations	Quantum Espresso + ASE	Complete
4	MEAM initialization & library merging	Python	Complete
5a	Composition NN surrogate (auxiliary)	TensorFlow + Keras	Complete
5b	NN-driven MEAM parameter optimization	TF + SciPy + LAMMPS	<i>Work in progress</i>

MD = Molecular Dynamics; DFT = Density-Functional Theory; NN = Neural Network.

Stage 1: Random Alloy Generator

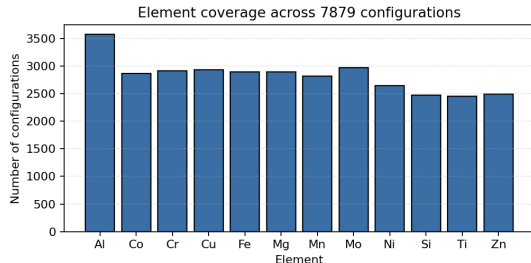


A Python generator draws compositions $\{c_i\}$ uniformly on the simplex $\sum_i c_i = 1$ over the 12 admissible elements, with **Al-dominant bias** (Al weight inflated when present) to oversample the engineering-relevant aluminium-alloy corner.

Output schema. Each composition is persisted as a structured record (element $\rightarrow$ mass-fraction map) at 10^{-4} precision, fully consumable by the downstream stages.

Coverage. ~ 7900 compositions generated; after the Stage-2 physicality filter, 7815 survive.

Element coverage



Element frequency over the surviving 7815 alloys; Al dominates by design.



LAMMPS [6]³ is the de-facto MD code in materials science: classical force-field integration on millions of atoms, MPI + OpenMP + GPU parallelism, scripted via an in-process Python API.

Why a custom LAMMPS build was necessary.

The upstream distribution caps the number of MEAM species at five, which is incompatible with the 12-element library produced in Stage 4. The species ceiling was therefore raised to twenty and a private build of the shared library installed alongside the standard tooling — *without this step the entire pipeline is blocked on the first configuration containing more than five distinct elements.*

Elastic protocol per configuration

- 1 Build ~ 4 nm cubic supercell from dominant element's lattice and composition-weighted a_0 .
- 2 Stochastically reassign atom types to match the requested composition.
- 3 Anisotropic minimisation (cell + positions).
- 4 Record baseline $\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$.
- 5 Apply $\delta = 10^{-3}$ axial strain ($\pm x, \pm y, \pm z$), re-minimise, record stress.
- 6 Central-difference $\rightarrow C_{11} \times 3, C_{12} \times 6$. Voigt average [7].

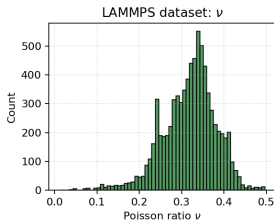
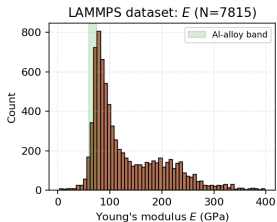
³LAMMPS = Large-scale Atomic/Molecular Massively Parallel Simulator (Sandia/Temple/NIST).

Stage 2: MD Results across ~ 7800 Alloys

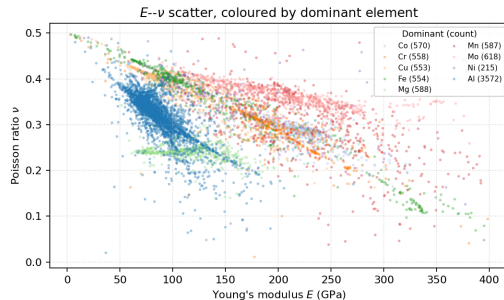


Quantity	N	Mean	Median	Std.	Min	Max
E (GPa)	7815	132.88	103.61	69.10	3.07	398.31
ν	7815	0.319	0.326	0.063	0.011	0.497

Dataset statistics after the $0 < E < 400$ GPa, $0 < \nu < 0.5$ physics filter.



Histograms of E (left) and ν (right). Green band marks the 60–75 GPa Al-alloy range.



E - ν scatter coloured by dominant element. Refractory-rich (Mo, Cr, Fe) populate the high- E wing.



Quantum Espresso [8]⁴ solves Kohn–Sham DFT [9] in a *plane-wave + pseudopotential* basis; QE 7.5 is driven through the **Atomic Simulation Environment** (ASE) [10] Python interface, so single-shot SCFs, EOS scans and elastic calculations are dispatched from a single process pool.

Computational settings

- Plane-wave cut-off: $E_{\text{cut}} = 40 \text{ Ry}$ (charge = 320 Ry)
- Marzari–Vanderbilt smearing [11], width 0.02 Ry
- Monkhorst–Pack mesh [12]: $6 \times 6 \times 6$
- Spin-polarised auto-on for Fe, Cr, Mn
- Pseudopotentials drawn from the official QE library on demand

Per-element computation

- ① Seven-point EOS scan $\rightarrow a_0, E_{\text{coh}}, B$ via **Birch–Murnaghan** fit [13, 14].
- ② Isolated-atom reference $\rightarrow$ true E_{coh} .
- ③ Strain protocol $\rightarrow C_{11}, C_{12}$ (same as LAMMPS).

Per binary pair. A single SCF on the $L1_2/B2$ reference structure $\rightarrow$ formation energy $E_{\text{form}}(i, j)$.

Validation. QE–ASE coupling sanity-checked on bulk FCC Al: $E \in [-540, -530] \text{ eV}$, $|F| < 0.01 \text{ eV}/\text{\AA}$, diagonal-only stress. All pass.

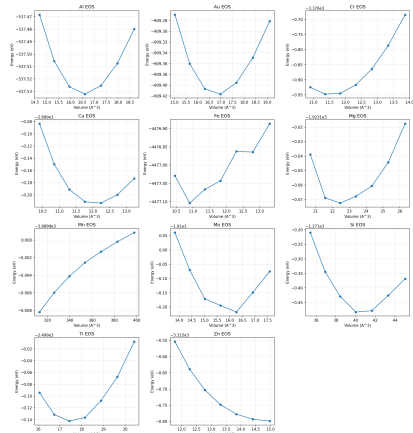
⁴QE = Quantum Espresso — open-source plane-wave DFT.

Stage 3: DFT Elemental References



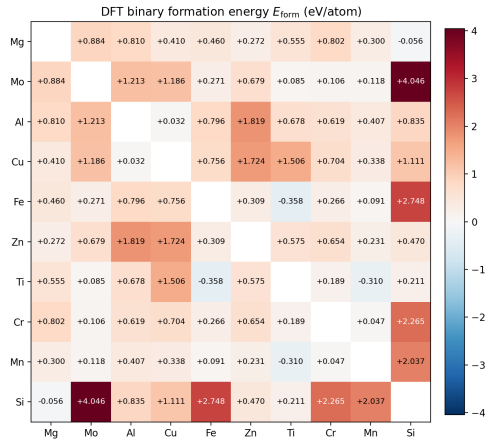
Element	Lat.	a_0 (Å)	E_{coh}	B (GPa)
Al	FCC	4.047	3.611	77.5
Cu	FCC	3.639	3.857	138.5
Si	DIAMOND	5.471	5.130	89.3
Ti	HCP	2.915	6.284	114.5
Zn	HCP	2.766	0.964	68.4
Cr	BCC	2.848	3.964	213.1
Fe	BCC	2.806	6.834	420.5
Mg	HCP	3.177	1.483	47.8
Mn	BCC	2.803	3.728	225.7
Mo	BCC	3.169	10.353	280.8

E_{coh} in eV/atom; B from Birch–Murnaghan fit to the seven-point EOS scan.



Birch–Murnaghan fits to seven-point QE EOS scans. The fitted minimum determines a_0 and E_{coh} ; the curvature determines B .

Stage 3: Binary Formation Energies



$E_{\text{form}}(i, j)$ in eV/atom from $L1_2/B2$ SCF references for the ten elements with a converged elemental calculation.

Negative (blue) $\Rightarrow$ exothermic mixing (e.g. Al–Ti, Cu–Ti — compatible *sd/sp* pairs).

Strongly positive (red) $\Rightarrow$ poor mutual solubility (notably Mg and Zn against most transition metals; Fe–Mo also stands out).

These signed values seed the cross-term $E_c(i, j)$ entries of the merged MEAM library directly, with no further fitting.

Stage 4: MEAM Single-Element Initialization



The MEAM screening exponent α is recovered from each element's DFT triple (a_0, E_{coh}, B) via

$$\alpha = a_0 \sqrt{\frac{9 B \Omega}{E_{\text{coh}}}},$$

where Ω is the primitive cell volume.

This populates a single-element MEAM block per element, which then feeds the cross-term refitter of Stage 5.

Mn caveat. The original EOS scan for α -Mn did not converge; values shown below are from a relaxed BCC approximation and the corresponding cross-terms inherit the higher uncertainty.

Element	Lat.	a_0 (Å)	E_{coh}	Ω (Å ³)	α
Al	FCC	4.047	3.611	16.57	4.470
Cu	FCC	3.639	3.857	12.05	4.930
Si	DIAMOND	5.471	5.130	20.46	4.473
Ti	HCP	2.915	6.284	35.02	5.987
Zn	HCP	2.766	0.964	29.93	10.921
Cr	BCC	2.848	3.964	11.55	5.906
Fe	BCC	2.806	6.834	11.05	6.179
Mg	HCP	3.177	1.483	45.36	9.063
Mn	BCC	2.803	3.728	11.01	6.120
Mo	BCC	3.169	10.353	15.91	4.924

E_{coh} in eV/atom.

Stage 5a: Composition $\rightarrow C_{ij}$ Surrogate (ML Sanity Check)



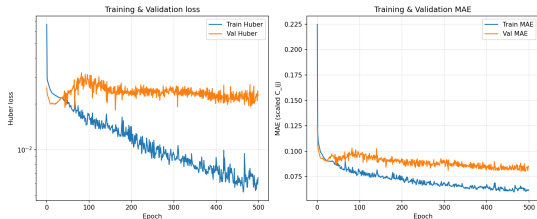
Role. A divergent ML branch that consumes the LAMMPS elastic database directly and never feeds back into MEAM fitting. *Two purposes:*

- Sanity-check the MD output — if a composition-only NN cannot recover (C_{11}, C_{12}) , the dataset itself is suspect.
- Deliver a fast predictor for novel compositions (< 1 ms per query).

Architecture. $12 \rightarrow 32 \rightarrow 20 \rightarrow 2$ MLP with ReLU; input is the 12-element mass-fraction vector $\{c_i\}$, output is $\hat{C}_{11}, \hat{C}_{12}$.

Loss. Huber ($\delta = 1$, scaled units) on C_{ij} — bounds outlier gradients from Mg/Zn-rich points.

Deep ensemble [15]. Five networks trained with independent seeds; the predictor returns mean $\pm$ std for epistemic uncertainty quantification.



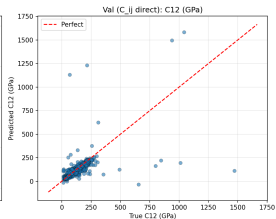
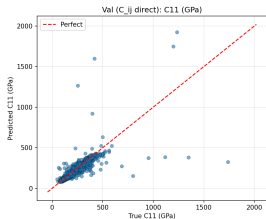
Per-epoch training history of one ensemble member. Huber loss (top) decays monotonically; MAE on (E, ν) (bottom) falls below the 20% mean-validation gate, which triggers early-stopping.

Stage 5a: Composition Surrogate Results (N=7839)

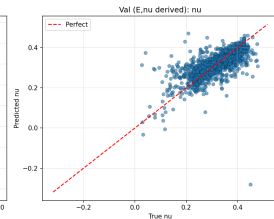
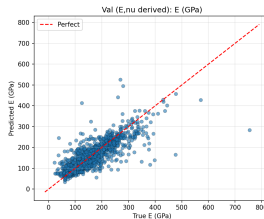


Split / Target	MAE	RMSE	Bias	R^2	r	MAPE
Train, C_{11} (GPa)	14.52	23.78	-5.51	0.967	0.985	7.14 %
Train, C_{12} (GPa)	10.09	18.81	-2.52	0.963	0.982	14.51 %
Train, E (GPa, derived)	18.04	28.18	-3.17	0.860	0.929	14.49 %
Train, ν (derived)	0.025	0.036	0.001	0.693	0.833	10.32 %
Val., C_{11} (GPa)	19.74	64.15	-5.46	0.646	0.822	8.56 %
Val., C_{12} (GPa)	14.32	57.32	-2.73	0.364	0.683	14.37 %
Val., E (GPa, derived)	20.44	34.98	-2.88	0.770	0.878	16.27 %
Val., ν (derived)	0.027	0.043	0.001	0.518	0.731	10.60 %

70 / 30 split: train = 5487, val = 2352.

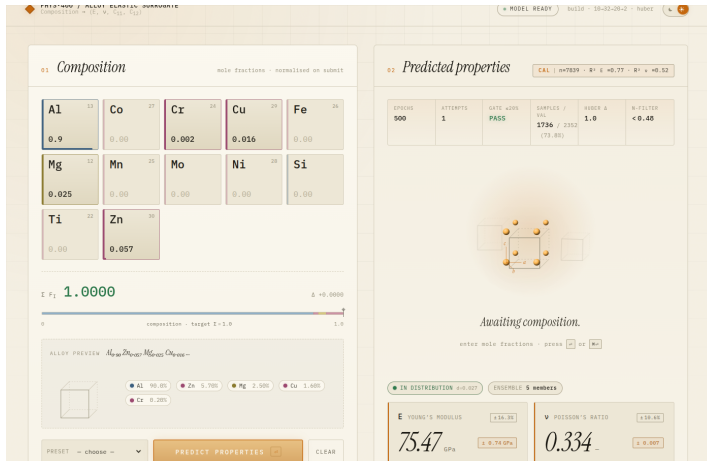


Held-out parity for C_{11} (left), C_{12} (right).



Derived (E , ν): $R_E^2 = 0.77$, $R_\nu^2 = 0.52$.

Stage 5a: Deployed Predictor — Interactive Composition UI



Browser-based predictor: the user enters element mass fractions, the trained deep ensemble returns $\hat{E}$, $\hat{\nu}$ with k-NN distance-based out-of-distribution flagging.

Validation on an AI 7075-like composition.

For the preset *AA7075-like* (Al–Zn–Mg–Cu–Cr) the surrogate returns

- $\hat{E} = 75.5$ GPa
- $\hat{\nu} = 0.334$

in line with the published AI 7075 range ($E \sim 71$ – 72 GPa, $\nu \sim 0.33$).

Distribution. Packaged as a portable ApplImage so the trained ensemble runs on any modern Linux desktop with no Python or CUDA toolchain installed.

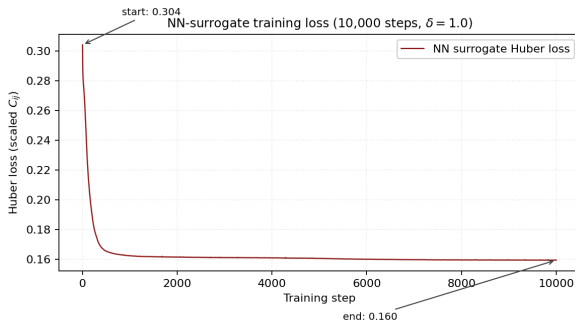


The closed loop.

- 1 A *differentiable NN surrogate* learns (MEAM params) $\rightarrow (C_{11}, C_{12})$ from a Sobol/active-learning sample of LAMMPS runs.
- 2 An L-BFGS optimizer [16] minimises $\sum_{\text{alloy}} \mathcal{L}_{\delta}(\hat{C}_{ij}, C_{ij}^{\text{MD}})$ *through the NN*.
- 3 Periodically: the optimised parameters are plugged back into LAMMPS, new MD samples appended, the surrogate re-trained.

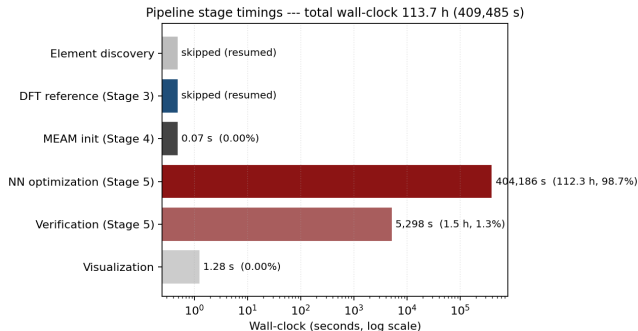
Training corpus. 70 / 30 train / val split of ~ 100 k-medoid representatives from the ~ 7800 -alloy database.

Resumable. Atomic checkpoint after every accepted LAMMPS sample; killing and re-running picks up where it stopped.



NN-surrogate training-loss decay over 10,000 steps ($\delta = 1$). The loss is on *scaled* C_{ij} — end value 0.16 corresponds to a residual surrogate RMSE of order 10–15 GPa on C_{11} .

Stage 5b: Where Did the 114 Wall-Clock Hours Go?

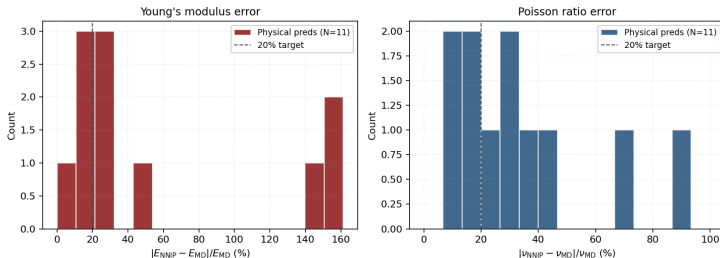


- **NN optimization dominates** (98.7 % of total wall-clock): every L-BFGS line-search evaluation triggers a full LAMMPS strain protocol over the 30 verification alloys.
- DFT and MEAM-init are effectively free at this scale — the cached DFT reference is reused, MEAM-init is 70 ms.
- This compute imbalance motivates an *active-learning sampler*: the next MD batch is selected by ensemble disagreement, rather than running LAMMPS blindly across a dense Sobol grid.

Stage 5b: Honest Verification on 30 Held-Out Alloys



NNIP Stage-5 verification on 30 held-out alloys | E within 20%: 4/30 | ν within 20%: 4/30 | unphysical NNIP outputs: 19/30



Quantity	Value
Held-out alloys	30
NNIP outputs passing the physicality filter	11
within 20 % of MD on E	4
within 20 % of MD on ν	4
Median E -error (physical subset)	$\sim 23\%$
Median ν -error (physical subset)	$\sim 26\%$

*The NNIP-optimized MEAM is, on the current 12-element corpus, **not yet a usable potential**: 19 of 30 held-out predictions fall outside the mechanical-stability region, and of the 11 that survive only ~ 4 are within the 20 % engineering target.*



Why does Stage 5b fail today?

- **Identifiability.** The cross-term block of a 12-element MEAM exposes ~ 200 free parameters against ≤ 100 medoid-selected composition targets — the inverse problem is under-determined.
- **Surrogate–LAMMPS mismatch.** The NN approximates the LAMMPS-elastic map with residual RMSE ~ 10 GPa; the L-BFGS optimizer chases artefacts of this residual into MEAM regions where the true LAMMPS gradient bends sharply.
- **Bound-saturation.** Several optimized parameters hit their box bounds (esp. $E_c(i,j)$, $\alpha(i,j)$), which is a tell-tale of an objective that wants to leave the physically meaningful region.

Planned next steps

- **Parameter-subset selection.** Freeze the single-element blocks (initialised by Stage 4), refit only the most-sensitive cross-terms identified by Sobol sensitivity.
- **Active learning expansion.** -sampling active already implemented; scale to 500+ medoids using ensemble-disagreement acquisition.
- **Physics-aware loss.** Add penalties for $\nu \rightarrow 0.5$ and $C_{11} < C_{12}$ *inside* the surrogate, not just at the verification gate.
- **Tighter QE coverage.** Add intermediate-composition SQS [17] DFT points so the cross-term loss can be evaluated against *multiple* reference compositions per pair.



Delivered

- **End-to-end pipeline** (Stages 1–4 + 5a): random alloy → MD elastic → DFT reference → merged 12-element MEAM library → composition surrogate.
- **Composition NN surrogate** with deep-ensemble uncertainty, $R^2 \sim 0.65$ on held-out C_{11} , $R^2 \sim 0.77$ on derived E , distributed as an interactive predictor.
- **Closed-loop NN-MEAM optimizer** (Stage 5b) with atomic checkpointing, three sampling regimes (Sobol, Latin hypercube, and active learning) and an honest verification protocol.

Methodological contribution

A closed-loop framework in which a *neural-network surrogate of the MEAM-to-stiffness map* is inverse-optimized against MD-measured targets — enabling MEAM construction from first-principles input alone, with no manual fitting and no experimental data. Verification of the NN-driven Stage 5b on the current 12-element corpus is the open problem to attack next.



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